

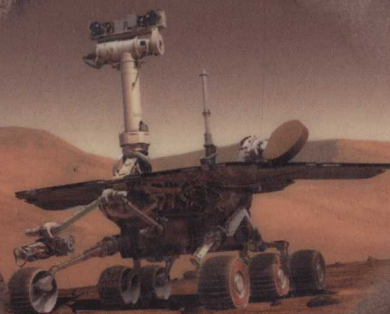
MARS

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Volume 2

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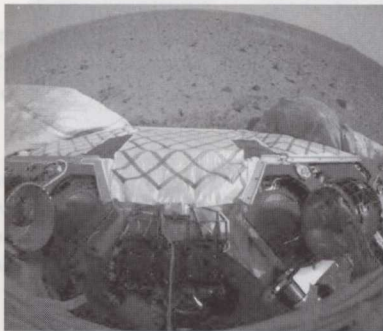


JPL, a division of the California Institute of Technology, manages the Mars Exploration Rover project for the NASA Office of Space Science, Washington.

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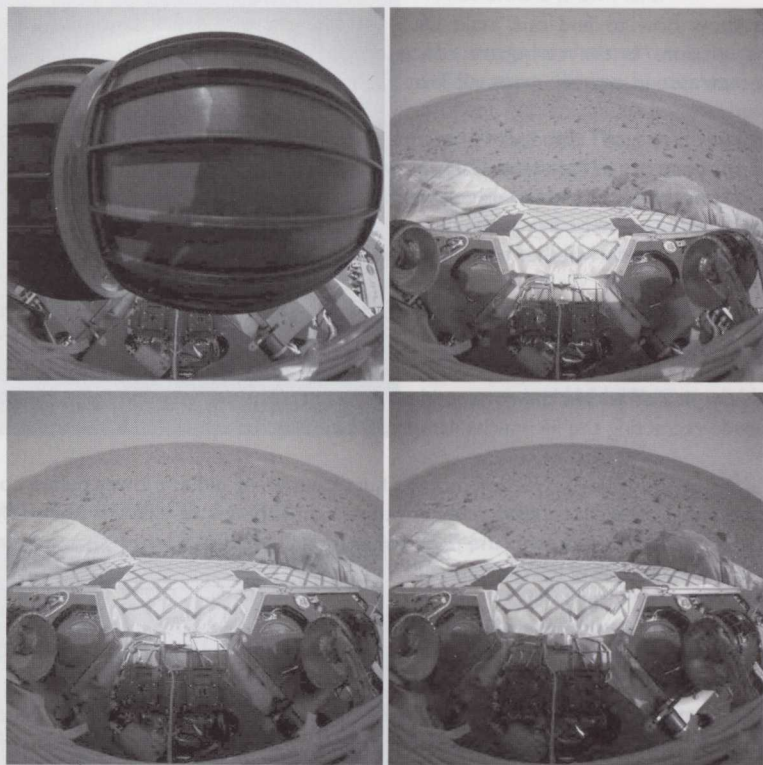
NEWS RELEASE: 2004-13



Spirit Rises to the Occasion

JPL engineers played Bob Marley's "Get Up, Stand Up" in the room as they watched new images confirming that the Mars Exploration Rover Spirit successfully stood up on its own. On Thursday night Pacific time, a major step in preparing for the next image from the rover's front hazard avoidance camera showed the rover in the final stage of its stand-up process. The two wheels at the bottom right and left are locked into position, along with the suspension system. The martian landscape is in the background. With a deflated airbag partially blocking one exit route, engineers were deciding whether Spirit should use a different route to roll off the landing site.

Image credit: NASA/JPL

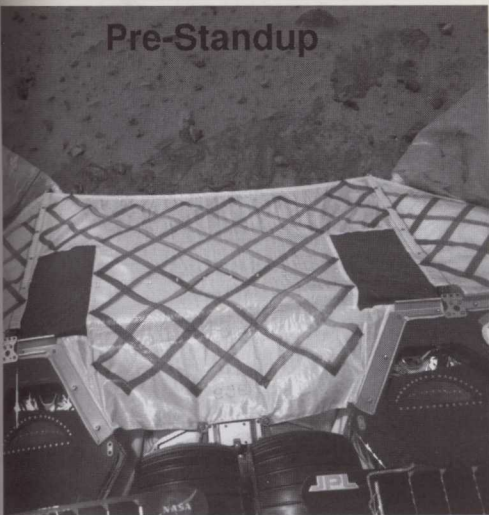


Spirit Rises to the Occasion (Animation sequence)

This animation strings together images from the rover's front hazard avoidance camera taken during the process of the Mars Exploration Rover Spirit. The first frame shows the rover's wheels tucked under it in its up position. The following frames show the stages of the stand-up process. The rover first elevates itself and then lowers the wheels. It then lowers, lifts and lowers again into its final position. Note the changing camera perspective of the martian landscape, indicating the rover's heightened and lowered positions.

Image credit: NASA/JPL

Pre-Standup

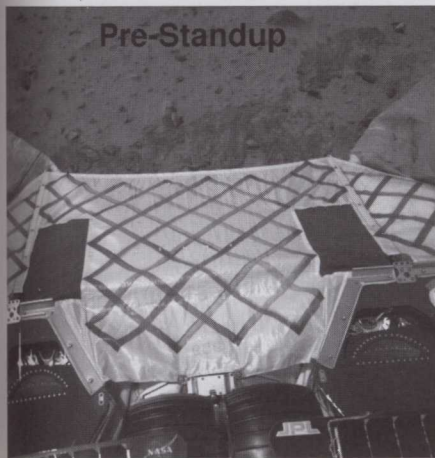


From Spirit's Perspective

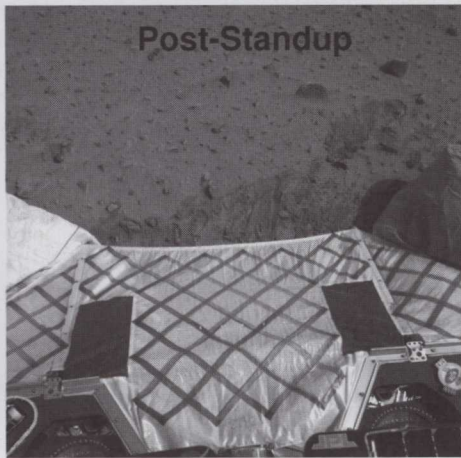
This is a perspective from the navigation camera on the Mars Exploration Rover Spirit prior to beginning the stand-up process.

Image credit: NASA/JPL

Pre-Standup



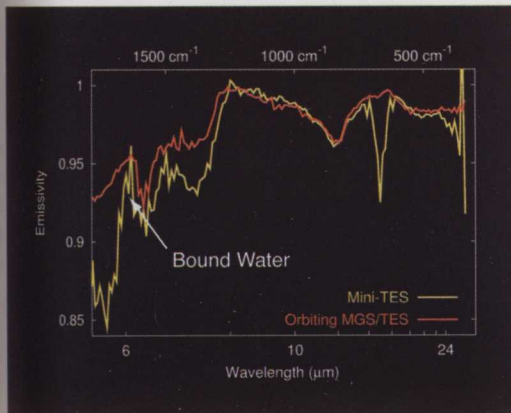
Post-Standup



From Spirit's Perspective (Animation sequence)

This animation shows the perspective from the navigation camera on the Mars Exploration Rover Spirit before and after its automated stand-up process. After standing up, the rover is approximately 12 inches higher off of the lander, resulting in a better view of the surrounding terrain.

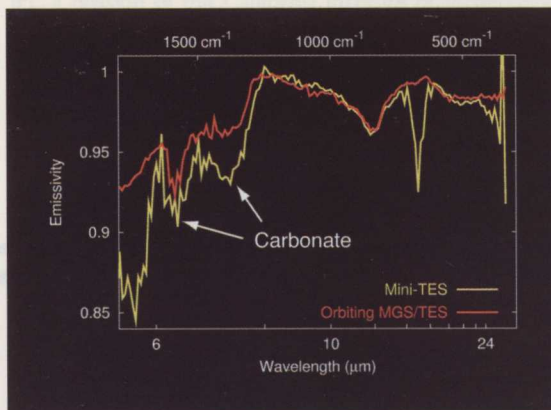
Image credit: NASA/JPL



Rover Senses Bound Water

This graph, consisting of data acquired on Mars from the Mars Exploration Rover Spirit's mini-thermal emission spectrometer, shows the light, or spectral, signature of an as-of-yet unidentified mineral that contains bound water in its crystal structure. Minerals such as gypsum and zeolites are possible candidates.

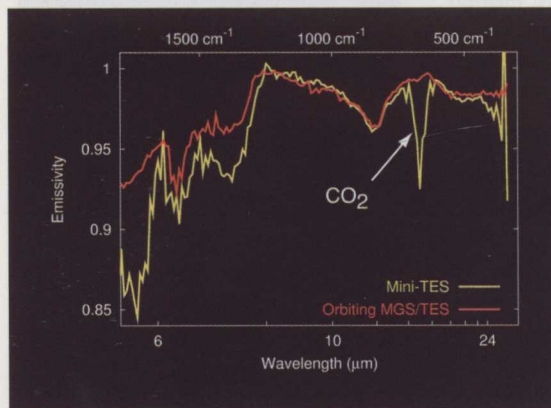
Image credit: NASA/JPL/Arizona State University



Rover Senses Carbonates

This graph, consisting of data from the Mars Exploration Rover Spirit's mini-thermal emission spectrometer, shows the light, or spectral, signatures of carbonates - organic molecules common to Earth that form only in water. The detection of trace amounts of carbonates on Mars may be due to an interaction between the water vapor in the atmosphere and minerals on the surface.

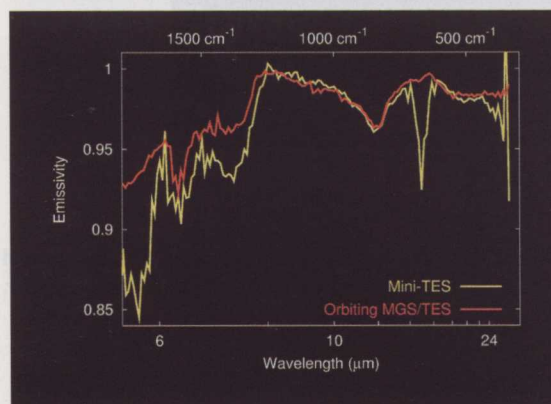
Image credit: NASA/JPL/Arizona State University



Rover Senses Carbon Dioxide

This graph, consisting of data acquired on Mars from the Mars Exploration Rover Spirit's mini-thermal emission spectrometer, shows the light, or spectral, signature of carbon dioxide. Carbon dioxide makes up the bulk of the thin martian atmosphere.

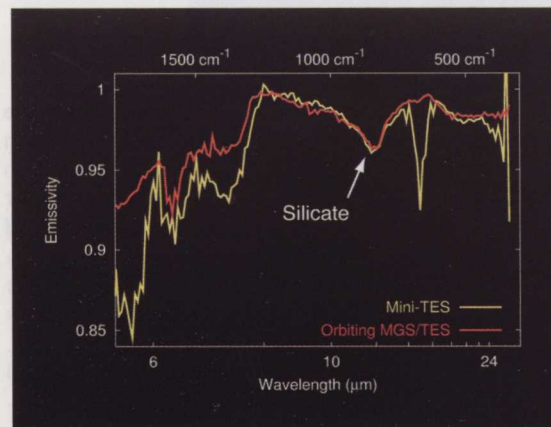
Image credit: NASA/JPL/Arizona State University



Dust Spectra from Above and Below

Spectra of martian dust taken by the Mars Exploration Rover Spirit's mini-thermal emission spectrometer are compared to that of the orbital Mars Global Surveyor's thermal emission spectrometer. The graph shows that the two instruments are in excellent agreement.

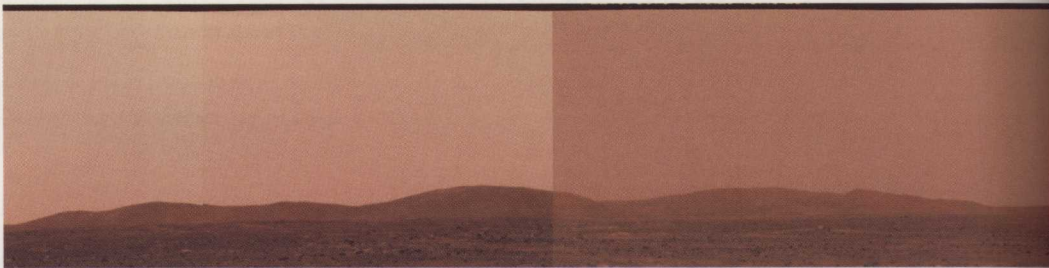
Image credit: NASA/JPL/Arizona State University



Rover Senses Silicates

This graph, consisting of data acquired on Mars by the Mars Exploration Rover Spirit's mini-thermal emission spectrometer, shows the light, or spectral, signature of silicates - a group of minerals that form the majority of Earth's crust. Minerals called feldspars and zeolites are likely candidates responsible for this feature.

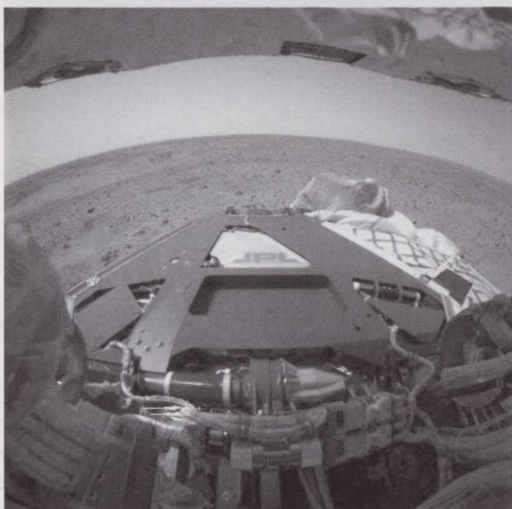
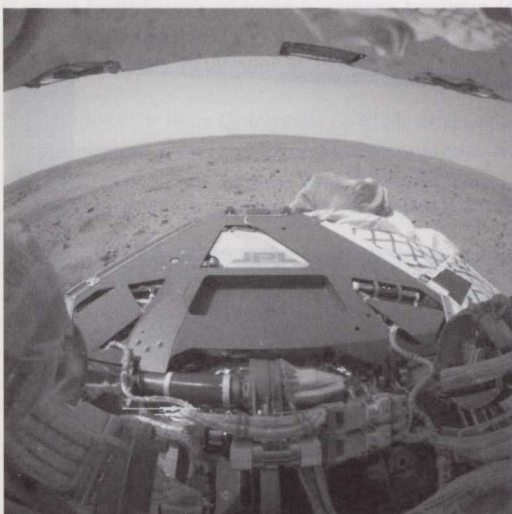
NASA/JPL/Arizona State University



Hazy Martian Skies

This image mosaic taken by the Mars Exploration Rover Spirit's panoramic camera shows the hills southeast of Spirit's landing site. Like a smoggy day in Los Angeles, dusty martian skies limit how much detail can be seen. Lack in visibility is demonstrated by comparing hills on the left to those on the right, located nearly two times farther away. The left panel of this image was captured in the late morning martian hours, looking toward the Sun. The image was taken in the early afternoon, when the Sun was higher and the skies appeared darker.

Image credit: NASA/JPL/Cornell University

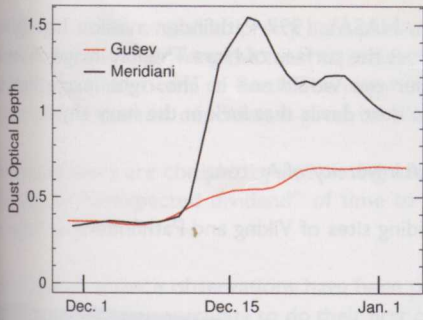


Spirit Stretches Out (Animation sequence)

This animation flips back and forth between images taken before and after deployment of the Mars Exploration Rover Spirit's bogie, a part of the rover's suspension system that extends the wheel base. These images were taken by Spirit's rear hazard avoidance camera.

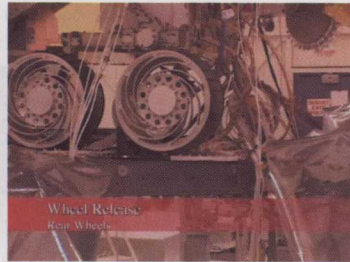
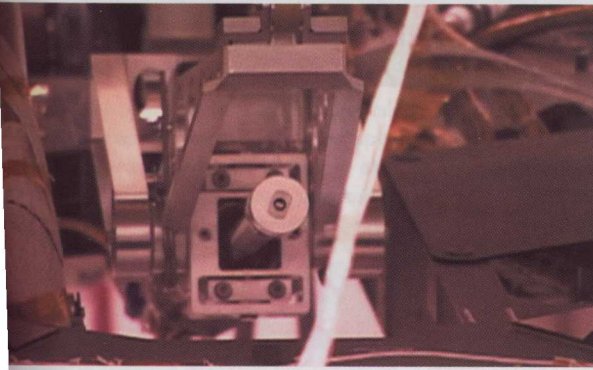
Image credit: NASA/JPL

Dust in the Wind

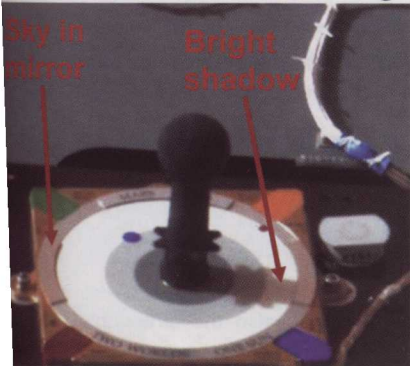


This plot shows the estimated change in dust levels from December 2003 to early January 2004 at Gusev Crater (red curve) and Meridiani Planum (black curve), the two Mars Exploration Rover landings sites. The measurements, retrieved from Mars Global Surveyor Thermal Emission Spectrometer, indicate that a large regional dust storm beginning in mid-December raised significant dust near Meridiani. Smaller amounts of dust were spread globally by winds, the effects of which were seen at Gusev Crater. For comparison, a dust optical depth value of 1.0 would correspond to a very smoggy day in Los Angeles or Houston, and a value of 0.1 to a relatively clear day in Los Angeles.

Image credit: NASA/Goddard/Arizona State University



Spirit color calibration target



True Colors of Mars

This image taken on Mars by the panoramic camera on the Mars Exploration Rover Spirit shows the rover's color calibration target, also known as the MarsDial. The target's mirror and the shadows cast on it by the Sun help scientists determine the degree to which dusty martian skies alter the panoramic camera's perception of color. By adjusting for this effect, Mars can be seen in all its true colors.

Image credit: NASA/JPL/Cornell University



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In 1877 the famed Italian astronomer Giovanni Schiaparelli used his brand-new 8.6 inch telescope to study the planets. To his great surprise he suspected that he saw symmetry on Mars. In the years that followed one astronomer after another looked at the red planet and gradually a mythology was formed—a mythology of alien intellect.

By the 1890's the martial influence had spilled over into all walks of life and sparked philosophical debates and wondrous fictions. Scientists, fantasists and people of all creeds looked up and wondered—is there life out there?

Now, more than a century later, nations around the world are bombarding Mars with an unprecedented fleet of exploratory vehicles. Their journey taking less time than it took Amundsen and Shackleton to reach the poles of Earth, these small but hardy robotic emissaries are thrusting their way through the depths of interplanetary space to take up residence in the barren Martian deserts. Their goal is to answer one of the oldest questions in mankind's history. Is there life out there?

In this sequel to the best-selling first volume, the reader is brought up to date with the most recent results from our nearest neighbour. Filled with a wealth of facts about the latest fleet of Martian explorers as well as a look at what may be coming next in mankind's most ambitious quest for knowledge.

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The Observatory

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